▣ Onboard Test Plan for Communication System Failure

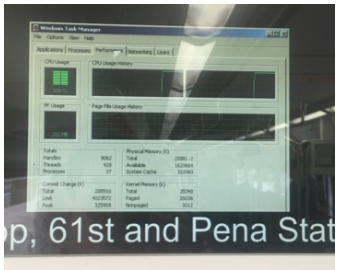
Issue

◈ Details

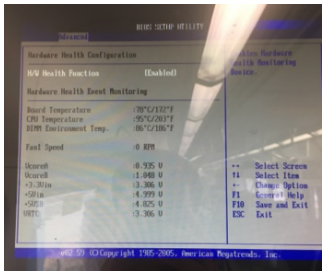
- The onboard test result for the communication system failure (Car 62, 57) during on July 5th ~ 6th in Denver.

1) Check CPU load status by an unknown reason.
2) Internal temperature rise by CPU overload. **(from 70 rising to 95 degree)**

3) Communication system function was affected by internal temperature rise.



<Overload Occurrence>



<CPU internal temperature rise>

◈ Cause

1) When GPS data is received through the network switch, the CNCS is overloaded due to the data proliferation by an unknown reason.

2) Failure will be continue to identify the cause during business trip.

Corrective action

◈ Temporary measure

- Restart the devices when CNCS is overloaded (Network data proliferation occurs)

◈ Action plan

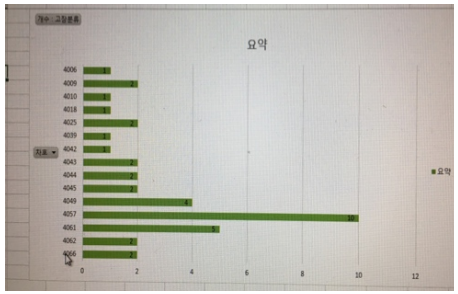
1) Install FAN at the top to forcibly cool the CNCS internal temperature

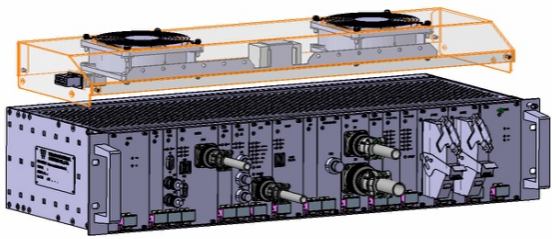
- The FAN is controlled by the temperature sensor and operated by sensing the temperature rise.

2) According to the failure analysis results from May to June, it is mainly occurred in specific car.

- Pre-correction of 10EA on the car (66,61,62,57,49,45,43,44,25,09) that the failure occurred over 2 times.

⇒ Plan : 7/10 ~ 7/14 (FAN installation and monitoring on 10 cars)





Microsoft Excel 워크시트

▣ Onboard Test Plan for Communication System Failure

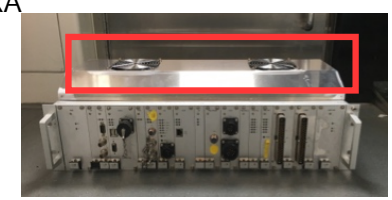
FAN Installation Method

◈ Installation Method

- On 7th July, FAN was installed on train 4035 and temperature change test was conducted.

1) FAN installation to the upper side of CRA

<Before FAN installation>



<After FAN installation>

2) Install CRA-CNCS temperature recorder

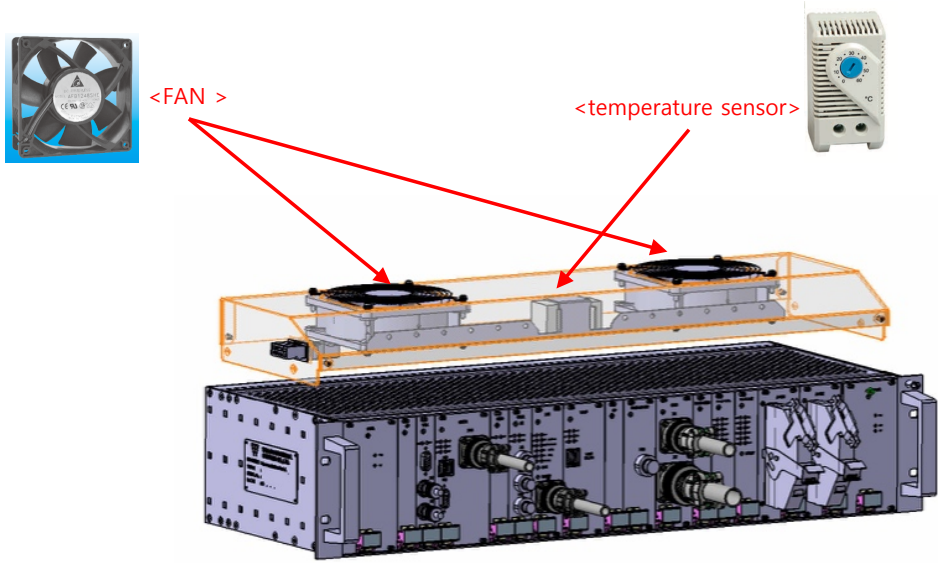
TR-71Ui

3) Install CRA on the car (FAN installation)

FAN Component Description and Operation Method

◈ FAN Operation Method

- Use train power as operating power
- The operating temperature can be set by the temperature sensor.
- The FAN operates when the CRA ambient temperature is higher than the set temperature.



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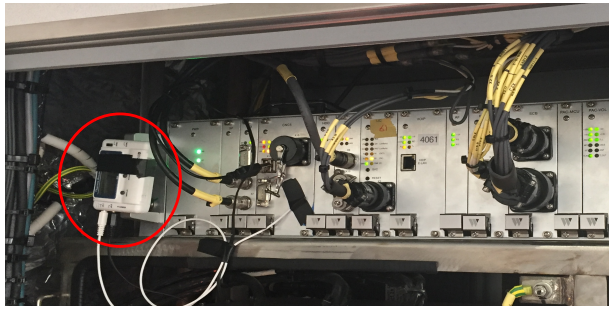
CRA Operating environment temperature

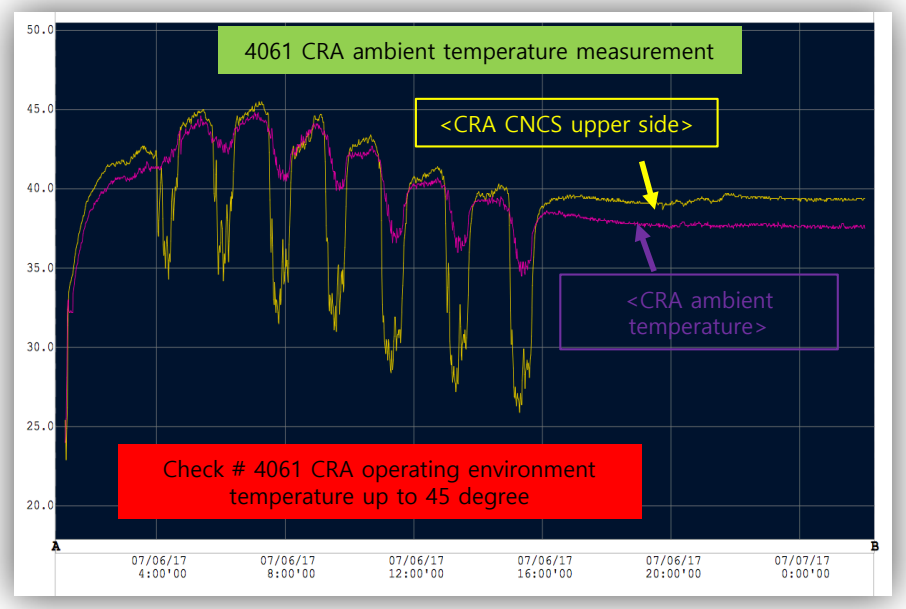
◈ Measurement of ambient temperature change

- CRA ambient temperature was measured on July 5-6 using temperature recorder (TR-71Ui)



TR-71Ui

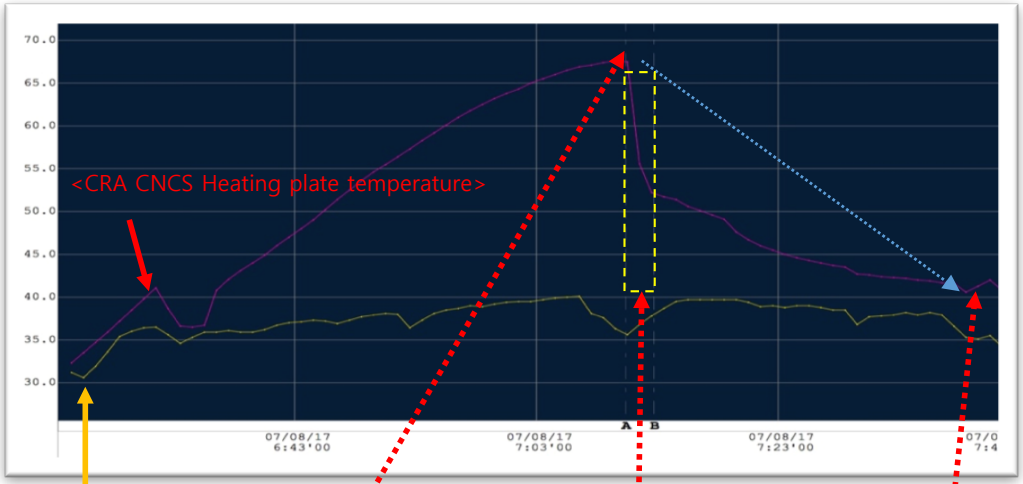
- CRA operating environment temperature up to 45 degree.



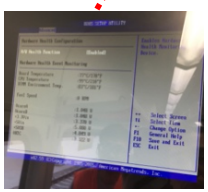
First onboard test result

◈ Test result

- When CPU overload is occurred by CPU monitoring, records the internal temperature.

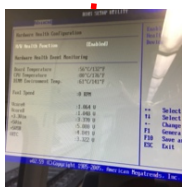


<CRA ambient temperature>



- CPU overload.
  → Communication system failure
- CPU internal temperature is 99 degree
- Heating plate is 67 degree

- FAN operation for 2 mins
  → Heating plate temperature is decrease from 67 to 50 degree
- Return the normal communication system function



- CPU internal temperature 80 degree.
- Heating plate is 40 degree

- After operating FAN for 2 mins, the communication system is returned to normal and decrease the ambient temperature.

- The ambient temperature is up to 45 degrees on the car.

Result: When FAN is operating, the temperature is decreased both